

Omar Hassan

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Alexandria, Egypt

I build models, benchmarks, and agents for science. My work spans protein design, biomedical research tooling, Egyptian Arabic evaluation, and efficient vision systems; contributor to Apple CoreML Tools and Kayba ACE.

EDUCATION

Alexandria University

Bachelor of Engineering in Biomedical Engineering — GPA: 3.2/4.0

Alexandria, Egypt

Expected Dec. 2027

EXPERIENCE

Made in Alexandria (M.I.A) Robotics

AI Engineer

Alexandria, Egypt

Sep. 2025 – Present

- Added SAM 3 / EdgeTAM video-backend support to X-AnyLabeling via `sam3.cpp`, enabling fast Apple Silicon annotation workflows and near-hands-free labeling of large video datasets on an M3 MacBook.
- Developed a training-free positive-unlabeled object detection system for FathomNetCLEF 2026 (CVPR-FGVC13 & LifeCLEF), detecting 32 marine species across 6,463 deep-sea images by combining Grounding DINO region proposals with DINOv3 k-means prototype matching and cosine-similarity background rejection; achieved mAP@[.50:.95] of 0.230 and ranked **#1** on the public leaderboard in the first week of competition.
- Built a real-time obstacle-awareness pipeline for an indoor assistive robot for visually impaired users, using YOLO with zone-based spatial reasoning and depth-aware priority scoring to classify obstacles by type, location, and urgency for TTS navigation alerts; reduced alert volume by 83.7% and improved inference throughput $5.7\times$ with a hybrid ONNX/MPS deployment architecture.
- Extended open-source evaluation tooling for Egyptian Arabic speech and translation, adapting Whisper benchmarking to MasriSpeech with Arabic normalization and WER/BLEU/RTF reporting, and adding Egyptian Arabic↔English LM Evaluation Harness tasks for NLLB-200 with BLEU, chrF, and TER.

Scientific Innovation Dynamics (SID)

Applied ML & AI Researcher

Alexandria, Egypt

Jul. 2024 – Sep. 2025

- Migrated a phonocardiogram processing pipeline for 8,000 recordings from SciPy scripts to a GPU-accelerated PyTorch dataset workflow on P100, cutting runtime from 5 hours to 40 minutes.
- Built ML systems for biomedical signal and imaging tasks, including a heart-sound classifier using MFCCs and a stain-free cell-detection pipeline trained on 1.8M+ phase-contrast microscopy images for microfluidic cell separation.

Esri North Africa

GIS & Geo-AI Intern

Cairo, Egypt

Jun. 2025 – Jul. 2025

- Fine-tuned the PRESTO satellite foundation model for multi-class crop classification across Africa and ranked **#37** globally on Zindi; built a GeoJSON/ArcGIS export pipeline to visualize predictions as feature layers in Esri workflows.

OPEN SOURCE

Apple CoreML Tools (*Python framework for ML model conversion*)

Open Source Contributor

github.com/apple/coremltools

2025 – Present

- Fixed unsafe graph optimization behavior for multi-layer LSTM conversion and implemented `l1_loss` as a composite MIL operation, improving model-conversion correctness and loss-function coverage. (PR #2645) (PR #2641)
- Improved quantization workflow documentation for developer clarity and onboarding. (PR #2640)

Kayba AI — Agentic Context Engine (ACE)

Open Source Contributor

github.com/kayba-ai/agentic-context-engine

2026 – Present

- Added an optional ACE MCP server with session-scoped tools, `stdio` transport, CLI packaging, safety guards, and test coverage, enabling external clients to use ACE through MCP. (PR #89)

SELECTED PROJECTS

HemoRL-Bench: Procedural benchmark for formula-verified coronary FFR reasoning, generating natural-language stenosis problems from a reduced-order pressure-drop model with unit-aware deterministic rewards; trained Nemotron-3-Nano-30B with GRPO, improving Tier 3 held-out accuracy from 38.8% to 60.0% and finding that outcome-only RL improved classification/dominant-loss reasoning while pressure-drop and inverse-flow tasks remained limited by errors in the viscous coefficient, which scales with stenotic diameter to the fourth power.

MedSci Agent: Open-source biomedical research agent exposing 28 MCP tools across drug discovery, protein analysis, omics, literature retrieval, medical imaging, and sandboxed execution, using MedGemma and TxGemma locally via Ollama with a long-running Python sidecar for scientific workloads.

PaperPing: Local-first iMessage research agent that ingests arXiv links and PDFs, answers follow-up questions with persistent 3-tier memory, supports deep paper reading and figure understanding, and sends automated morning briefings using Bun, SQLite, FTS5, and multi-step LLM tool calling. Product website.

HireWorkflow AI: Multi-agent hiring workflow for technical recruiting that jointly evaluates resumes and GitHub repositories, orchestrating candidate parsing, code-aware validation, scoring, and downstream workflow automation to reduce manual screening time and ATS-style false negatives.

Simple Protein Agent: Protein design agent built with Google ADK, Gemini, and Apple SimpleFold that generates candidate sequences from natural-language specifications and predicts 3D structures; engineered a sequential ESM-3B → SimpleFold pipeline that reduced peak memory from about 24 GB to about 12 GB.

SKILLS

Languages: Python, TypeScript, SQL

ML/AI: PyTorch, ONNX, Computer Vision, Speech/ASR, Arabic NLP, Multimodal & Agentic Systems

Systems: FastAPI, Docker, SQLite, Git

Scientific Computing: Scanpy, RDKit, BioPython